Experiment-4

Capacitance-Voltage Characteristics of PN Junction Diode

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Aim: Design silicon PN junction diode and study the C–V characteristics using Synopsys TCAD Sentaurus.

Objectives:

Study junction capacitance vs. voltage and analyze the depletion capacitance.

Software Tool:

Synopsys TCAD Sentaurus.

Procedure:

1.Create a new folder. Open Terminal in the new folder then enter “sde &” command in the terminal.
2.Go to View a GUI Mode a 2D Mode.
3.Go to Draw then deselect the “Auto Region Naming “and select the “Exact Coordinates”.
4.Now go to materials and select “silicon” (silicon is the default material) then click on “Create rectangular Sheet Region” on the toolbar.
5.Place the rectangle on Canvas with (0,0.045) and (0,0.180) coordinates and press ok.
6.Now enter Region Name as “nregion”.
8.Now create the p region inside the n region for that select Snap to Grid from Tool Bar.
9.Select the Rectangle, draw the p region from the top left corner of n region with below mentioned coordinate values.
10.For doping the nregion, Go to Device à Constant Profile Placement. Select the region as nregion and change the concentration as required.
11.For doping the pregion, Go to Device Constant Profile Placement. Select the region as pregion and change the concentration as required.
12.For creating Contact, Go to Contacts Contact Sets. Enter the contact name for pregion and nregion with different edge colour.
13.To Set Contact for pregion and nregion first change the Select Body to Select Edge on Tool Bar.
14.Now click on Arrow placed at left panel and go to Contacts Set Contact then place the electrical contact on the edge of pregion and nregion.
15.Go to Mesh à Define Rel/Eval Window and select rectangle à draw rectangle on canvas covering the entire design.
16.Got Mesh à Refinement Placement enter the minimum & maximum element size.minimum size is 1e-06 and 0.003 is the maximum element size and Click on create Refinement.
17.Go to Mesh click on Build Mesh. Then the device structure will be open in svisual.
18.To study the device characteristics, edit the diode “ac cmd file” (.cmd file). msh.tdr file name should be included in cmd file Grid section. Change the file section as mentioned in the below figure and the frequencies varying from 100Hz, 1000Hz and10000.
19.Enter the ac file “sdevice cmdfilename” (Ex: sdevice TEST_ac.cmd) in working directory terminal.
20.To visualize the device characteristics, enter svisual command in terminal.
21.Go to file in svisual and open plot file(pndiode_CV_100Hz_ac_des.plt).

22.For plotting C-V characteristics select pcontact(Vg) on x-axis and Cg-g to Y- axis.

RESULT:

Fig 1:concentration of PN junction diode

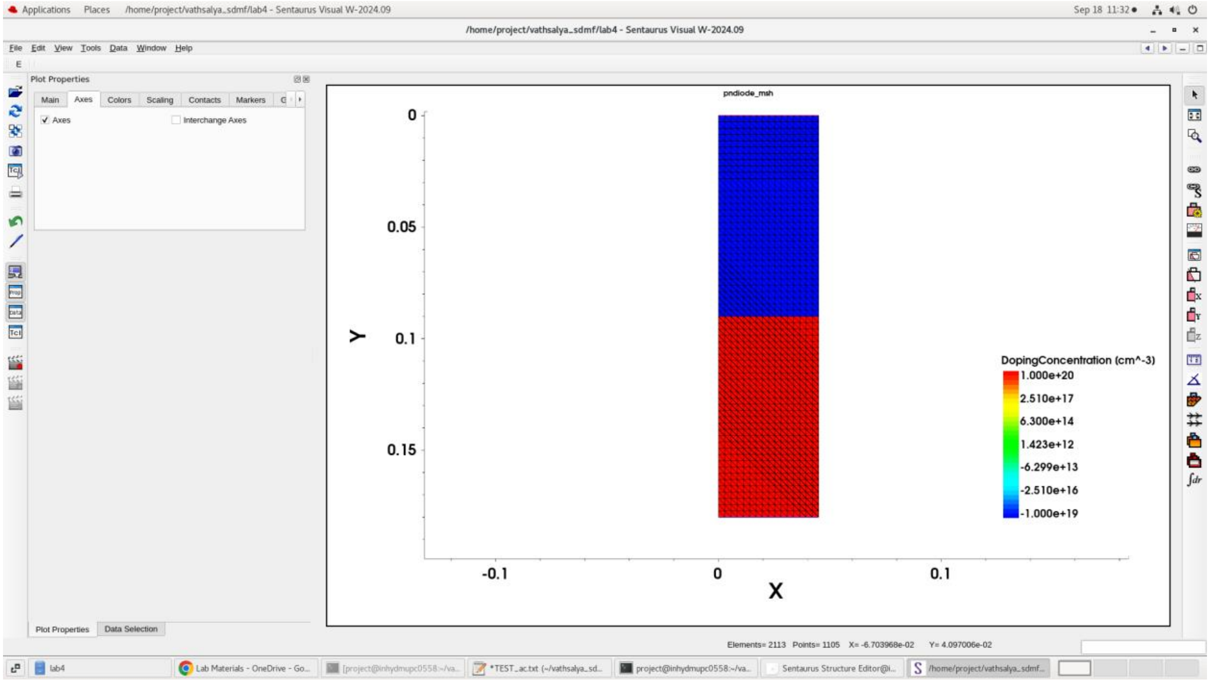
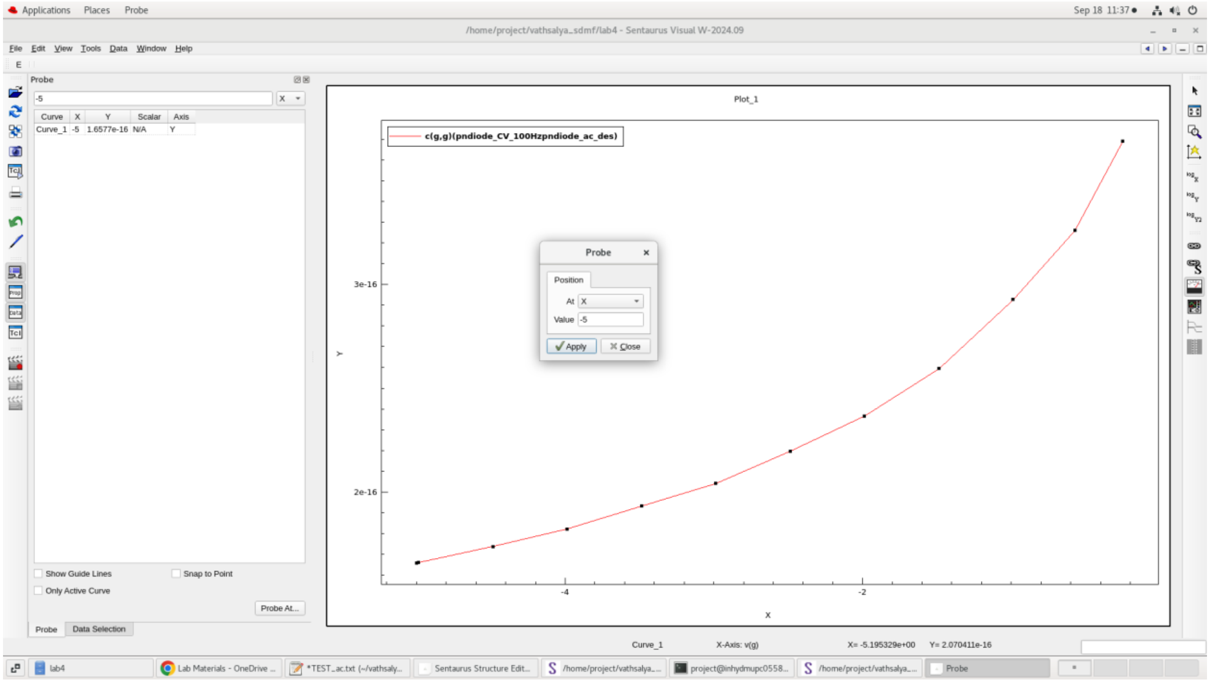


Fig 2:C-V characteristics of PN diode where concentration are 1e+19 and1e+20 respectively

2a:At -5v



2b:At -0.25v

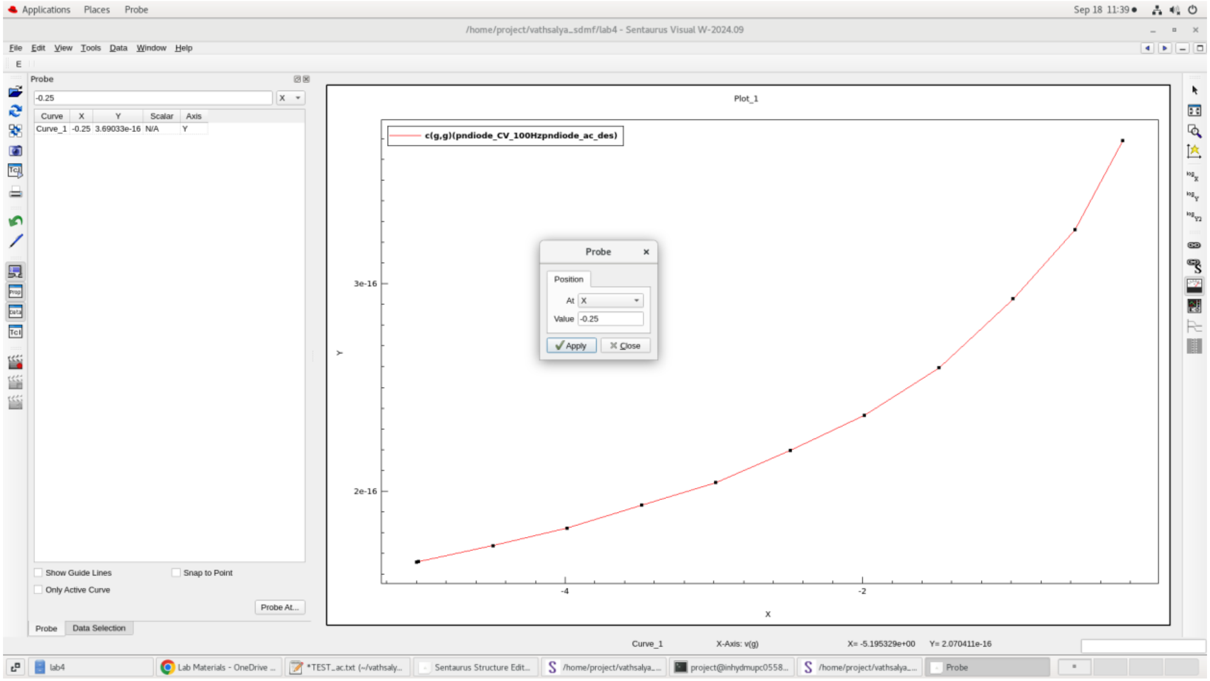


Fig 3:C-V characteristics of PN diode where concentration are 1e+15 and1e+17 respectively

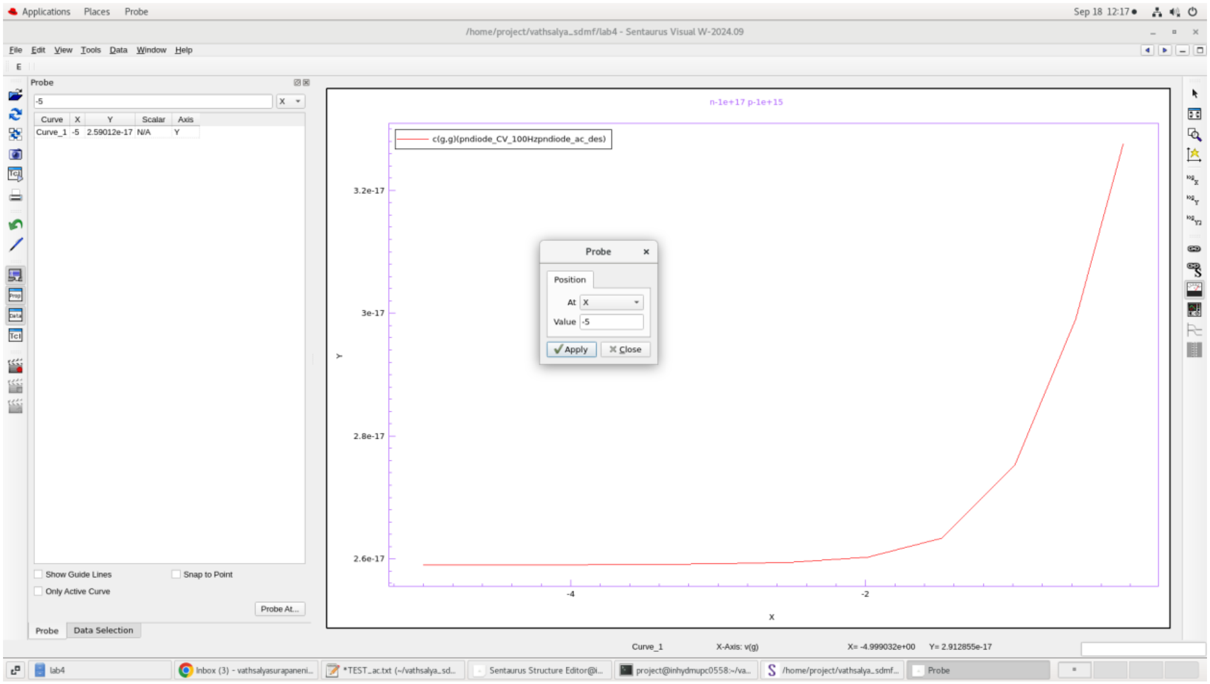
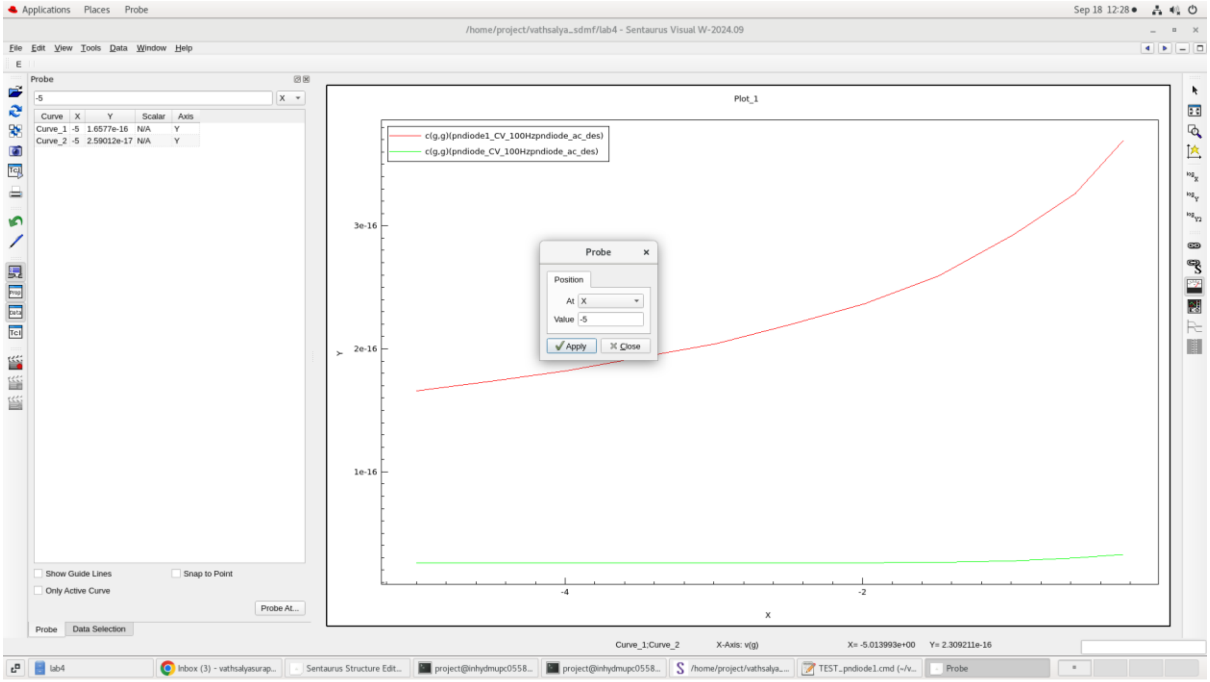


Fig 4: change in capacitance with change in concentration



OBSERVATION:

1:Capacitance reduces as the voltage reduces from 0 to -5v.

The capacitance at -0.25v is 3.69033e-16 and the capacitance at -5v is 1.6577e-16.

CONCLUSION:

The C-V characteristics of PN junction diode are studied using Synopsys TCAD Sentaurus.